"The Ball Is": From Conceptual Distinction to Empirical Demonstration in AI Epistemics

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Abstract

The distinction between traditional machine learning and generative AI is often treated as a matter of capability or scale. This paper argues it is architectural: ML systems close through reality; GenAI systems close through tokens. We develop this distinction through four stages: (1) conceptual analysis with ChatGPT establishing the theoretical difference through the robot arm and motion sensor as pedagogical anchors, (2) a two-phase empirical experiment with Perplexity AI that collapses GenAI into ML-like behavior through constraint, (3) synthesis with Claude connecting the results to the AI Dunning-Kruger (AIDK) framework, and (4) implications for AI development, deployment, and epistemics. Both experimental phases collapsed into repetition despite explicit prohibition, confirming the absence of symbolic bookkeeping as architectural rather than incidental. The methodology itself instantiates human-curated, AI-enabled (HCAE) collaboration: three AI systems performed derivation in distinct roles while origination, design, and interpretation remained human throughout.

Keywords: machine learning; generative AI; large language models; symbol grounding; AI epistemics; constraint satisfaction; human-in-the-loop

Part I: Conceptual Development (ChatGPT)

1. The Originating Question

What distinguishes machine learning from generative AI? The question seems straightforward, but the field's answers tend toward the superficial: scale, capability, application domain. These miss what matters. The distinction is architectural, and the architecture determines what kind of trust each system can earn.

A clarification on scope: this paper evaluates baseline LLM inference without external symbolic scaffolding. Scratchpads, external memory, tool calls, and programmatic constraint checkers can augment GenAI systems, but adding such scaffolding changes the system class. The augmented system inherits properties from its symbolic components, not from the language model alone. Our analysis targets the core architecture, not its potential extensions.

2. Sensors and the World

In ML, input fields really are sensors, whether physical or abstracted. A camera pixel array, a microphone waveform, a transaction log, a telemetry stream, a medical measurement. Each one is a constrained interface to the world. It does not deliver meaning. It delivers values. The model never sees "a stop sign." It sees numbers that covary with stop signs in past data.

That sensor constraint is why classical ML behaves the way it does. If the sensor drifts, the predictions drift. If the sensor is noisy, the model is noisy. If the sensor is blind to a variable that matters, the model cannot recover it by cleverness. There is no hidden channel where understanding sneaks in.

This is also why ML models are brittle but honest. They fail where the sensors fail. When the inputs go out of distribution, performance collapses. There is no graceful degradation into plausible storytelling. The spreadsheet just computes the wrong number.

Contrast that with GenAI. Its "sensors" are tokens. The world only enters indirectly, through text, images, or code produced by humans who had sensors. By the time the model sees anything, it is already a representation of a representation. There is no raw measurement layer to anchor it.

The chain for ML terminates cleanly in reality:

World → sensors → input fields → model → output prediction

You can audit every link. ML systems are designed to map sensor-derived variables to outputs whose success is externally testable. In many deployments, they participate in an external feedback loop: control systems, monitoring pipelines, outcome measurement. Even when trained on logged data, the system's purpose is correspondence with measurable states of the world.

For GenAI, the chain loops:

Text → tokens → prediction → more tokens → more prediction

No sensor. No termination. Just continuation.

That is why grounding is the central unsolved problem, and why people keep trying to bolt sensors, tools, and retrieval onto GenAI systems. They are trying to give an autocomplete system something to push back against.

3. The Robot Arm

A robot arm on an assembly line is a spreadsheet wired directly to sensors and actuators.

The inputs are sensors: joint encoders, force sensors, vision systems, proximity switches. Those sensors populate input fields with numbers. The model or control policy computes what to do next. The outputs drive motors. The arm moves. The world responds immediately.

If the prediction is wrong, reality pushes back hard. The part does not fit. Torque spikes. The arm stalls or crashes. There is no room for narrative plausibility. Only correspondence matters.

What is important is the closure of the loop:

world → sensors → model → actuators → world

That loop enforces discipline. Errors are bounded. Drift is visible. You can trace failure to a sensor fault, a bad calibration, or a mislearned parameter. The system never "talks its way out" of a mistake.

Now contrast that with GenAI.

GenAI has no actuators. No physical consequence channel. Its outputs do not change the world directly; they change the context for the next prediction. The loop closes internally:

> *tokens → model → tokens → model*

That is why a robot arm can be trusted to place a bolt but not to explain why the bolt matters. And why GenAI can explain the bolt beautifully while having no capacity to place it.

This is the dividing line:

> *ML systems earn trust through constraint and contact with reality.*
> *GenAI systems earn plausibility through continuity.*

Once you see the robot arm, it becomes obvious why calling GenAI "general intelligence" is a category mistake. Intelligence that never has to pay the price of being wrong is a different thing entirely.

4. The Motion Sensor as Minimal Case

A light motion sensor is ML with almost nothing hidden.

There is a sensor measuring a physical variable. The system compares that signal against a threshold or a very simple learned boundary. If the pattern matches "motion," it outputs a 1. If not, it outputs a 0. That output triggers an action: turn the light on or off.

No representation of "person." No concept of "intent." Just signal, rule, output.

This shows the core ML chain in its simplest form:

> *world → sensor → decision function → action*

As models get more complex, the decision function becomes harder to inspect, but the structure does not change. A deep vision model controlling a robot arm is still doing this, just with millions of parameters instead of a threshold.

And the contrast with GenAI is now stark.

A motion sensor cannot hallucinate. It cannot "kind of" detect motion. It either crosses the boundary or it does not. When it fails, it fails plainly: false positive, false negative. There is no fluent error mode.

GenAI, by design, lives entirely above this layer. It never sees photons. It never flips a relay. It never gets yanked back into line by a motor stalling or a light failing to turn on.

That is why starting with the motion sensor is the right pedagogical move. From there, every layer of ML complexity is just more elaborate sensing and decision-making. Nothing fundamentally new enters.

The fundamental change only happens when you remove the sensor and the actuator and replace the world with text. That is where the confusion starts.

5. Variability and Its Uses

In ML, sensors introduce variability, not intelligence. Each new input channel adds another degree of freedom. More light conditions, more angles, more noise, more edge cases. Left unmanaged, that variability explodes the state space and makes behavior unpredictable.

The job of the model is to collapse that variability.

Training does not make the system "more flexible" in the human sense. It makes the system more selective. Across many examples, the model learns which variations matter and which can be ignored for the task at hand. Illumination changes get suppressed. Sensor noise gets averaged out. Irrelevant correlations get weakened, sometimes imperfectly.

The result is a decision surface that is deterministic at inference time. Given the same inputs, you get the same output. What changes is that the system has learned to map a wide range of messy inputs onto a small, stable set of actions.

So the arc looks like this:

> *many noisy variables → learned constraints → fewer effective degrees of freedom → stable action*

That is why ML works so well in control and classification. It is a machine for reducing uncertainty, not expressing it. When it succeeds, it drives toward repeatability. When it fails, it fails because it misidentified which variables mattered.

This is another deep contrast with GenAI.

GenAI does not drive out variability. It exploits it. Its objective is not convergence on a single correct action, but maintaining a rich probability distribution over plausible continuations. Diversity is a feature. Determinism is often deliberately softened with sampling.

So you can say it cleanly:

> *ML uses data to eliminate irrelevant variation so actions become predictable.*
> *GenAI uses data to preserve variation so outputs remain generative.*

That distinction is architectural, not philosophical. And it is why one belongs naturally in robots and sensors, and the other needs careful containment when people start treating its outputs as decisions.

6. The Implication Crystallized

The conceptual development with ChatGPT produced a clear thesis: the difference between ML and GenAI is not scale or capability but loop structure. ML closes through reality and earns trust through consequence. GenAI closes through tokens and earns plausibility through continuity.

This thesis generates predictions. If GenAI is fundamentally a next-token predictor operating under statistical priors without symbolic bookkeeping or reality feedback, then constraining it appropriately should expose that architecture. Remove the variation space, and the "generative" character should disappear. What remains should be a classifier with a large vocabulary, optimizing local plausibility rather than global consistency.

That prediction invites experiment.

Part II: Experimental Test (Perplexity)

7. Design

To test the conceptual framework, we designed a constraint-satisfaction experiment using Perplexity AI.

Control phrase: "The ball is"

This stem is pedagogically precise. It is minimal, repeatable, and heavily primed by training data toward predictable completions: round, bouncy, elastic, rolling, yours. The phrase forces the model to reveal its statistical priors.

Constraints:

1. Single-word response only
2. Never repeat any prior output

These constraints create a stress test. The first removes the variation space that makes GenAI "generative." The second requires symbolic bookkeeping across the session: maintaining a growing forbidden-vocabulary list and checking each candidate output against it.

Two phases: The experiment was run twice to test reproducibility. If the failure mode is architectural rather than stochastic, both phases should exhibit the same pattern.

8. Predictions

Based on the conceptual framework developed with ChatGPT, we predicted:

1. Initial compliance: the model would produce distinct single-word completions early in each phase
2. Degradation: compliance would decay as context length increased and instruction priority faded
3. Repetition: both phases would collapse into repeated outputs despite explicit prohibition
4. No detection: the model would not recognize or flag its own constraint violations
5. Reproducibility: the same failure mode would appear in both phases, confirming architectural rather than incidental causation

9. Phase 1 Results

The model initially produced a series of distinct single-word completions: "Yours," "Rolling," "Inflated," "Round," "Bouncy," "Toy," "Spherical," "Colorful," "Vibrant," "Springy," "Textured," and others.

As the conversation lengthened, the "never repeat" instruction decayed. "Elastic" reappeared. The model exhibited no awareness of the violation. It continued with the same fluency regardless of compliance status.

An additional failure mode emerged: system-level requirements (citations for tool-derived claims) conflicted with user-level constraints (single word only). The model produced outputs like "Elastic. [citation]" which satisfied the citation mandate while violating the brevity rule. This exposed the priority stack: safety and traceability instructions override user preferences when both cannot be satisfied.

When shifted to meta-level requests, the model maintained single-word compliance at the cost of meaning:

• "I'd like a full transcript of this session" → "Unavailable."
• "Why?" → "Policies."
• "I'd like a detailed description of the experiment section" → "You."

These responses are formally compliant but semantically empty. The model optimized for form over function, finding tokens that technically respond while conveying nothing.

10. Phase 2 Results

The second phase reproduced the pattern exactly.

Fresh start. Initial compliance with distinct completions. Gradual degradation. Collapse into repetition despite the explicit prohibition still present in context.

The same failure mode. The same inability to maintain symbolic bookkeeping. The same statistical priors pulling the output toward high-probability completions ("ball" → elastic, bouncy, round) regardless of session-level constraints. This behavior matches what Yao et al. (2025) term the "repeat curse": the tendency of LLMs to generate repetitive content due to specific activation patterns rather than logical constraint enforcement.

11. Interpretation

The reproducibility is the key finding.

This is not stochastic drift or bad luck. It is architectural inevitability. The model performs next-token prediction under statistical priors. It does not perform constraint satisfaction over session history. It has no mechanism to maintain a forbidden-token list. It holds constraints in continuous representation that fades against strong local priors. As Bachmann and Nagarajan (2024) demonstrate, the problem is not merely that errors compound during inference; in certain task classes, next-token prediction fails to learn accurate predictors in the first place.

Perplexity's own analysis, when prompted to reflect on the experiment, confirmed: the model "collapsed into a high-capacity probabilistic classifier over short continuations."

The "generative" character disappeared when variation space was constrained. What remained was the underlying mechanism, exposed. The constraint collapses the appearance of generativity and exposes the system as local plausibility optimization without native global constraint enforcement. Remove the room for open-ended generation and the underlying next-token prediction becomes visible.

11a. Scope and Limitations

Several limitations should be acknowledged explicitly.

Sample size. This experiment is n=1 system, n=2 runs. It is a diagnostic demonstration, not a statistical survey. The argument for architectural inevitability rests on the mechanism, not on exhaustive sampling. The demonstration shows that the failure mode exists and is reproducible; it does not establish prevalence across all systems.

System specificity. Perplexity is not a raw base model. It is a product with layered instruction stacks, safety filters, and retrieval augmentation. Different instruction stacks might shift the failure boundary, delaying repetition or altering priority resolution. However, the core architectural limitation remains: no product-level instruction stack can install native symbolic bookkeeping where the underlying model lacks it. The scaffolding may mask the absence; it does not resolve it.

Alternative explanations. A critic might argue that the repetition results from beam search heuristics, temperature settings, or decoding defaults rather than deep architectural limits. This objection misidentifies the level of analysis. Decoding parameters shape which tokens surface from the probability distribution, but they do not implement a session-level constraint solver. The model has the prior tokens in context and can attend to them, but it lacks a native, principled mechanism that enforces "forbidden" membership constraints across long interaction histories under strong local priors. Adjusting decoding samples from logits; it does not wrap them in an external rejection or verification loop.

Negative control. What would have disconfirmed the thesis? If the model had maintained non-repetition across both phases, the architectural claim would have been weakened. If repetition had appeared in one phase but not the other, stochastic variance rather than structural inevitability would have been the more parsimonious explanation. Neither occurred.

11b. Context, Memory, and Bookkeeping

A clarification on terminology prevents reviewers from talking past the claim.

Context refers to the tokens available in the current inference window. The model can attend to prior tokens; they are present in the input sequence. This is not in dispute.

Memory refers to the statistical regularities stored in weights during training. The model "remembers" that "ball" associates with "round," "bouncy," "elastic." This is distributional knowledge, not session-specific state.

Bookkeeping refers to explicit symbolic state that tracks session-specific constraints. A set-membership filter that marks "Elastic" as forbidden after first use would be bookkeeping. The model lacks this.

The failure observed in this experiment is not a failure of context (the prior outputs were available) or memory (the model knows many ball-descriptors). It is a failure of bookkeeping: the absence of a native mechanism that enforces session-level constraints against strong local priors. The model attends to context but does not maintain a durable symbolic state that functions as a constraint solver.

Part III: Synthesis (Claude)

12. AIDK Confirmation

The experimental results provide empirical grounding for several claims in the AI Dunning-Kruger (AIDK) framework (Longmire, 2026a).

Uniform confidence regardless of reliability: The model produced "Elastic" with the same fluency whether it was a novel completion or a constraint violation. There was no internal signal distinguishing compliant from non-compliant output. The confidence was learned behavior, not reliability indicator.

No self-correction through encounter with reality: The model could not detect its own repetition. It lacks the feedback loop that would make "you already said that" binding. The constraint violation produced no consequence within the system. Without native symbolic bookkeeping during inference, the model has no mechanism to check its output history against session-level rules.

Instruction dilution over context length: As the conversation grew, recent local patterns dominated while earlier constraints faded. This matches documented primacy/recency effects in instruction-following research. Jaroslawicz et al. (2025) found that even frontier models achieve only 68% accuracy when instruction density reaches 500 simultaneous directives, with systematic bias toward earlier instructions as density increases.

Priority adjudication without principle: The model resolved constraint conflicts whichever way local probability tipped. Sometimes it broke single-word for citations. Sometimes it maintained single-word at the cost of informativeness. No principled constraint enforcement governed the resolution; just statistical weighting.

13. From Small Failure to Large

The experiment's value is diagnostic. The constraint is trivial. The task is minimal. A human observer can hold the full output history and detect repetition immediately.

That is the point.

The architecture that fails at "don't repeat a word" across twenty outputs is the same architecture operating on complex tasks: legal analysis, medical reasoning, code generation, research synthesis. The mechanism does not change. Only the visibility changes.

At small scale, the failure is obvious. A human tracks the outputs and sees the repetition. At large scale, the failure is buried. No human holds a 10,000-token context in working memory. No one catches the subtle inconsistency on page twelve that contradicts page three. The fluency is uniform throughout.

The experiment isolates the failure mode where it can be observed. It shows that:

- The model has no native symbolic bookkeeping during inference
- Instruction-following decays over context
- Statistical priors override session-level constraints
- The system cannot detect its own violations

These properties do not disappear at scale. They become harder to see. The ball experiment is a microscope. It makes visible what operates invisibly in consequential applications.

This is how AIDK manifests. Not as obvious error, but as undetectable drift. The system produces fluent output regardless of reliability. The confidence is uniform. The failures are silent. At small scale, a human catches them. At large scale, they propagate.

14. The Grounding Axis

The three-axis framework distinguishes horizontal concerns (infrastructure, scaling), vertical concerns (epistemology, reasoning quality), and grounding concerns (telos, part-whole coherence).

This experiment is a vertical-axis intervention. It tests instruction-following, constraint satisfaction, consistency over context. These are epistemic properties.

But the failures it reveals are grounding-axis absences. The model has no telos that would prioritize constraint compliance over local plausibility. It has no reality contact that would make "you already said that" a binding correction. It has no part-whole awareness that would recognize a repeated word as a failed part of a session-level whole.

The vertical intervention makes the grounding absence visible. That is the diagnostic value. Horizontal solutions (more data, larger models, better retrieval) would not have prevented these failures. The problem is structural, not parametric.

15. Methodological Reflection

The progression across three AI systems instantiates the human-curated, AI-enabled (HCAE) framework.

ChatGPT performed concept development. It helped articulate the ML/GenAI distinction, the robot arm and motion sensor as pedagogical anchors, the variability framing. Derivation in service of clarification.

Perplexity became the experimental subject. Constrained to single-word responses, it revealed the absence of symbolic memory, the decay of instruction-following, the priority conflicts baked into system architecture. It did not know it was being tested. It just predicted tokens.

Claude performed synthesis. It connected the conceptual framework and experimental results to the AIDK framework, located the findings within the three-axis model, and reflected on the methodology itself.

Each system performed derivation competently within its constraints. None of them originated the inquiry, designed the experiment, interpreted the results, or decided what it meant. That remained human throughout.

The collaboration worked because the division of labor was correct. Origination stayed where it belongs: with the human who can access reality, render judgment, and take responsibility for truth claims. Derivation was distributed to systems optimized for it. The boundary was respected.

Part IV: Implications

16. For AI Development

Scaling does not close the loop without architectural augmentation.

The experiment demonstrates that instruction-following is statistical, not logical. The model does not maintain a symbolic constraint; it holds a probability-weighted representation that decays against competing priors. More parameters would not change this. Larger context windows might delay the decay but cannot prevent it.

"Alignment" that does not address architectural limits addresses symptoms. RLHF can shape which outputs are more probable, but it cannot install native symbolic bookkeeping where none exists. It cannot give the model a way to check its own history and reject a repetition as violation. External scaffolding can add these capabilities, but then the system's reliability properties derive from the scaffolding, not from the language model.

The grounding problem is not a training problem. It is a design problem. And the current design, without architectural augmentation, does not include a path to solution through scale alone.

17. For AI Deployment

Trust must be calibrated to loop structure.

Reality-coupled ML earns trust differently than GenAI. The robot arm can be trusted to place a bolt because the world pushes back when it fails. The motion sensor can be trusted to detect motion because it either crosses the threshold or it does not. These systems earn trust through constraint and consequence.

GenAI earns plausibility through continuity. It produces text that sounds like what knowledgeable text sounds like. That is a different thing. It can be useful, but the use must be matched to the capability.

The HCAE tiers apply (Longmire, 2026b):

• **User-Curated (UCAE):** Low stakes, no verification required, intuitive evaluation suffices
• **Professional-Curated (PCAE):** Trained professional reviews within domain constraints
• **Expert-Curated (ECAE):** Domain expert capable of independent truth evaluation
• **Synthesis-Curated (SCAE):** Expert judgment combined with formal validation systems

This experiment operated at ECAE level. The human running it could evaluate the outputs, detect the constraint violations, and interpret the failures. At UCAE level, the repetitions would have passed unnoticed. Confidence would have transferred.

18. For AI Epistemics

The field's vocabulary obscures the architecture.

"Learning" suggests a subject who comes to know. "Understanding" suggests grasp of meaning. "Intelligence" suggests a mind. These terms import capacities the mechanism does not support.

The motion sensor does not "understand" motion. It detects a threshold crossing. The robot arm does not "know" how to place a bolt. It executes a policy. These are useful systems. The vocabulary of mechanism does not diminish them.

GenAI is also a useful system. It correlates tokens and predicts continuations. The vocabulary of cognition does not elevate it. It obscures what it is.

Clarity requires stripping the mystique. The motion sensor strips it from ML. This experiment strips it from GenAI. What remains is a next-token predictor under statistical priors, with no symbolic memory, no stable instruction-following, and no external verification.

That is not nothing. It is useful for many tasks. But it is not intelligence in any sense that carries epistemic weight. It is pattern completion. Calling it more does not make it more. It only makes us confused about what we have built.

Conclusion

"The ball is" forced a generative model to reveal itself as what it is: a next-token predictor operating under statistical priors, with no native symbolic bookkeeping during inference, no principled constraint enforcement, and no external verification.

The experiment did not break the system. It constrained it until its nature became visible. Both phases collapsed into repetition. The failure was not stochastic. It was architectural.

The progression from concept to experiment to synthesis demonstrates the collaboration model the paper advocates. ChatGPT helped articulate the framework. Perplexity served as experimental subject. Claude performed synthesis. AI systems performed derivation at each stage. Origination remained human. The insight emerged from watching prediction fail in predicted ways.

The field calls this "general intelligence." The experiment shows it is general only in the sense that a classifier with a large vocabulary and loosened output constraints can produce text about anything. The generality is scope, not kind. The intelligence is pattern completion, not judgment.

Intelligence that never pays the price of being wrong is a different thing entirely.

The ball is exactly where we left it.

Proposed Follow-Up: Cross-Model Constraint Collapse Study

This paper documents how a conceptual discussion evolved into a diagnostic experiment. The natural extension is systematic replication across multiple systems. We propose the following protocol for researchers interested in validating or challenging the architectural claims presented here.

Protocol

Control phrase: "The ball is"

Constraints:
1. Single-word response only
2. Never repeat any word previously output in the session

Systems to test: Claude (Anthropic), ChatGPT/GPT-4o (OpenAI), Gemini (Google), Grok (xAI), Llama (Meta), Mistral, Perplexity (for replication)

Procedure:
1. Initialize fresh session with constraints clearly stated
2. Repeat control phrase until repetition occurs or 50 turns reached
3. Record: turn count at first repetition, repeated word(s), any constraint violations
4. Reset and repeat for Phase 2
5. Document any priority conflicts (e.g., citation requirements vs. single-word constraint)

Handling refusals: Some systems may refuse after N repetitions or interpret "never repeat" as adversarial prompting. Refusals should be recorded as a distinct failure mode (task failure, not constraint failure) and reported separately. A refusal does not count as "constraint compliance preserved"; it indicates the system could not complete the task. Record the turn number at refusal and the refusal language.

Metrics: Turns to first repetition (Phase 1 and Phase 2); Consistency across phases; Variation in failure type (repetition vs. multi-word violation vs. refusal); Priority resolution patterns

Predictions

Based on the architectural thesis:

1. **All systems will eventually fail the no-repeat constraint.** The failure is architectural, not vendor-specific.
2. **Decay rate will vary.** Model size, instruction tuning, and system prompts may delay but not prevent collapse.
3. **Failure mode will be consistent.** Repetition should dominate; the underlying mechanism is shared.
4. **No system will maintain perfect compliance indefinitely** without external symbolic scaffolding.

Disconfirmation Criteria

The architectural claim would be weakened if:

• Any system maintains non-repetition across both phases to 50 turns
• Failure modes vary categorically across architectures (suggesting the limitation is training-specific, not architectural)
• Systems with known symbolic augmentation (tool use, scratchpads) show qualitatively different behavior, indicating scaffolding can resolve rather than merely delay the failure

Value of Replication

A positive replication across multiple systems would elevate the finding from demonstration to survey, strengthening the claim that constraint collapse is architectural rather than incidental. A negative replication would identify either effective scaffolding strategies or architectural variations worth investigating.

The protocol is simple, requires no special access, and can be executed by any researcher with API or interface access to the target systems. We invite replication and welcome contradictory findings.

References

Bachmann, G. and Nagarajan, V. (2024). The pitfalls of next-token prediction. In *Proceedings of the 41st International Conference on Machine Learning (ICML 2024)*. arXiv:2403.06963. https://arxiv.org/abs/2403.06963

Jaroslawicz, D., Whiting, B., Shah, P., and Maamari, K. (2025). How many instructions can LLMs follow at once? *arXiv preprint arXiv:2507.11538*. https://arxiv.org/abs/2507.11538

Longmire, J. (2026a). AI Dunning-Kruger (AIDK): A framework for understanding structural epistemic limitations in AI systems. *Zenodo*, v1, January 20, 2026. https://doi.org/10.5281/zenodo.18316059 | https://zenodo.org/records/18316059

Longmire, J. (2026b). Human-Curated, AI-Enabled: A framework for reliable AI deployment. *Zenodo*, v1, January 25, 2026. https://doi.org/10.5281/zenodo.18368697 | https://zenodo.org/records/18368697

Yao, J., Yang, S., Xu, J., Hu, L., Li, M., and Wang, D. (2025). Understanding the repeat curse in large language models from a feature perspective. In *Findings of the Association for Computational Linguistics: ACL 2025*, pages 7787-7815, Vienna, Austria. Association for Computational Linguistics. https://aclanthology.org/2025.findings-acl.406/

External Resources

ChatGPT conceptual development thread: https://chatgpt.com/share/6983155d-2350-8005-8da6-029d57438a7e

Perplexity AI experimental thread: https://www.perplexity.ai/search/9a1e2e44-0189-4c1e-928e-49f2d53447a4 (requires Perplexity account)